ROHIT RAKSHAN PALANIKUMAR

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EXECUTIVE SUMMARY

Aerospace engineering senior with experience developing propulsion analysis tools, CFD simulations, aerospace structures, and integrated hardware systems in fast-paced research and production environments. Background includes propulsion modeling, turbulent flow analysis, composite structures, and mechanical integration through internships, CubeSat development, and Formula SAE aerodynamic design.

EXPERIENCE

The Boring Company – *Integration Engineering Intern* **September 2025 – December 2025**

- Supported assembly and integration of 3+ Prufrock TBMs across multi-disciplinary builds
- Managed MBOMs across 5 active projects with 500+ parts; corrected design errors to reduce integration rework
- Partnered with production techs and test engineers to build and validate mechanical assemblies hands-on

Flamefront Propulsion – *Engineering Intern* **June 2025 – August 2025**

- Built turbojet engine modeling tool in Python from scratch using GitHub libraries, ingesting Mach, altitude, and geometry inputs to output thrust, SFC, and required propulsion performance metrics across the flight envelope
- Implemented component selection logic with mechanical and thermodynamic constraints; achieved 6% thrust increase
- Designed engine geometry in CAD from component analysis outputs, closing analysis-to-hardware loop

Computational Fluid Physics Laboratory – *Undergraduate Researcher* **October 2024 – August 2025**

- Investigated programmable microjet flow control in turbulent boundary layers to extend attached flow over NACA 4412
- Ran high-fidelity CFD cases on TACC supercomputer using Nek5000; validated Cl/Cd against XFOIL achieving 93% prediction match via span increase
- Microjet made a 42% decrease in separation volume for fast LSMs; indicating aerodynamic performance improvement

ORGANIZATIONS

Longhorn Racing Electric (FSAE Team) – *Aerodynamics Engineer* **August 2023 – Present**

- Designed flip-ups and bargeboard in SolidWorks; generating 40N downforce and 16N drag
- Optimized sidepod geometry to hit radiator cooling inlet flow targets via ANSYS and Cadence CFD
- Manufactured aero components using carbon fiber layups and resin infusion; reinforced with aluminum ribs and structural spars

Texas Spacecraft Laboratory

Communications Test Engineer **February 2024 – May 2025**

- Conducted thermal-vacuum testing of spacecraft electronics across -50°C to +100°C and 10^{-3} Pa, documenting results and validating component performance under flight-representative environmental conditions
- Performed trade studies of S-Band ground station components to meet mission link requirements for 10cm CubeSat supporting NASA ARTEMIS GPS positioning

Lead Design Engineer

- Led FEA-based structural optimization of 1.25kg sensor payload (8×8×5 in) under 10N load case, ensuring structural integrity and integration readiness across subsystems

LRA Vertical Active Stabilization – *Avionics Bay Sub-Team Lead* **August 2023 – March 2024**

- Led design and testing of fin geometries for active vertical stabilization, iterating configurations in SolidWorks and validating stability margins in OpenRocket
- Collaboratively designed and built avionics bay to house flight electronics, ensuring structural fit and component accessibility for integration and recovery

PROJECTS

TLOFT DARPA Heavy Lift Challenge

- Designed VTOL airframe CAD targeting 160 lb payload capacity on a 40 lb frame (4:1 lift-to-weight ratio) across a 7 ft wingspan and 39×39 in quadcopter frame
- Performed Euler buckling analysis to validate structural integrity at FOS of 5 for primary load-bearing members
- Simulated VTOL trajectory via rotor dynamics modeling, resolving optimal cruise AoA of 14° at 18 m/s for heavy-lift flight envelope

EDUCATION

Bachelor of Science in Aerospace Engineering (Honors) – University of Texas at Austin **August 2023 - December 2026**